NASDAQ Index Forecasting

ECON 5140: Applied Econometrics

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Objective

Construct ARIMA, ETS, and Prophet models to forecast future NASDAQ Performance

Observe data from 2016-01-04 to 2026-02-20

Create a forecast of the next 5 trading days

Compare models accuracy on a 5-day, 1-month, and 1-year time horizon back testing against actual data

ARIMA

1. Transform the data → **2. Select model order (p, d, q)** → **3. Diagnostic check** → **4. Forecasting**

1

- NASDAQ prices show a clear upward trend: non-stationary series.
- We difference the data to remove the trend and obtain a stationary series for ARIMA.

2

- ARIMA(2,1,0). Chosen by the lowest AIC (Akaike Information Criterion).
- d=1: one differencing applied. p=2: captures short-term dependence between days.

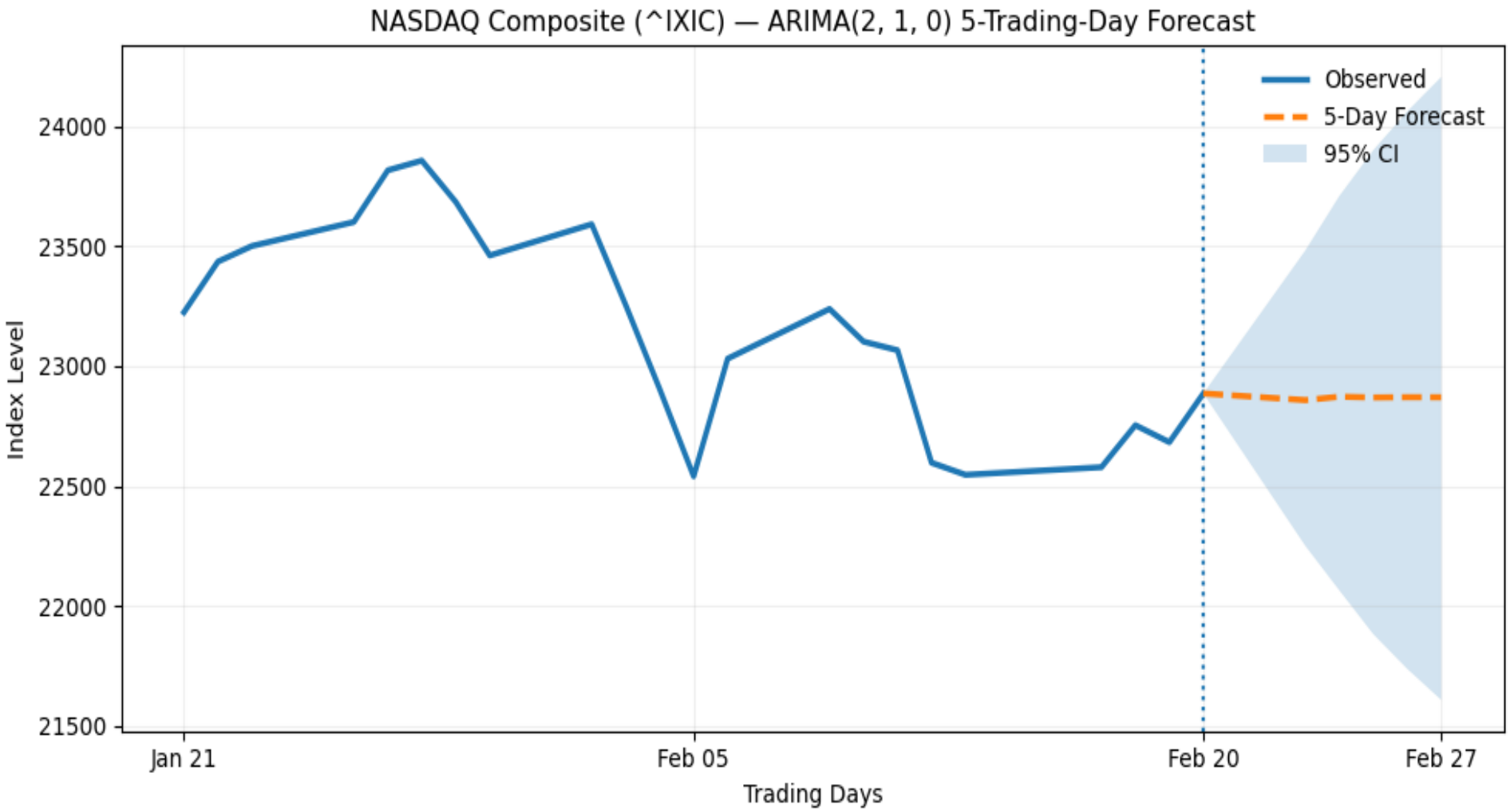
3

- Ljung–Box test shows no autocorrelation (p-value ≈ 1.0).
- Residuals behave like white noise → model is adequate.

4

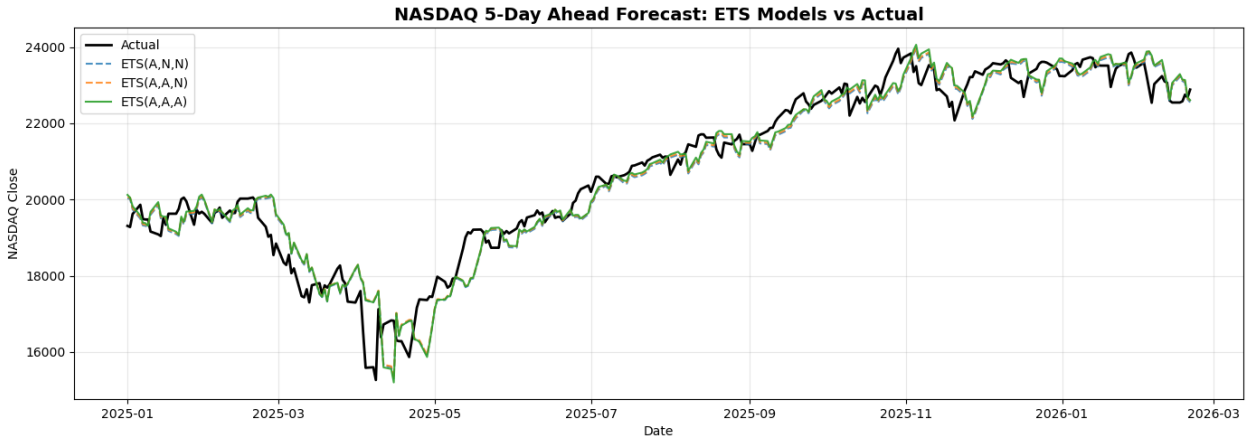
- Predict the next 5 trading days.
- Closer days are more reliable; uncertainty increases farther into the future.

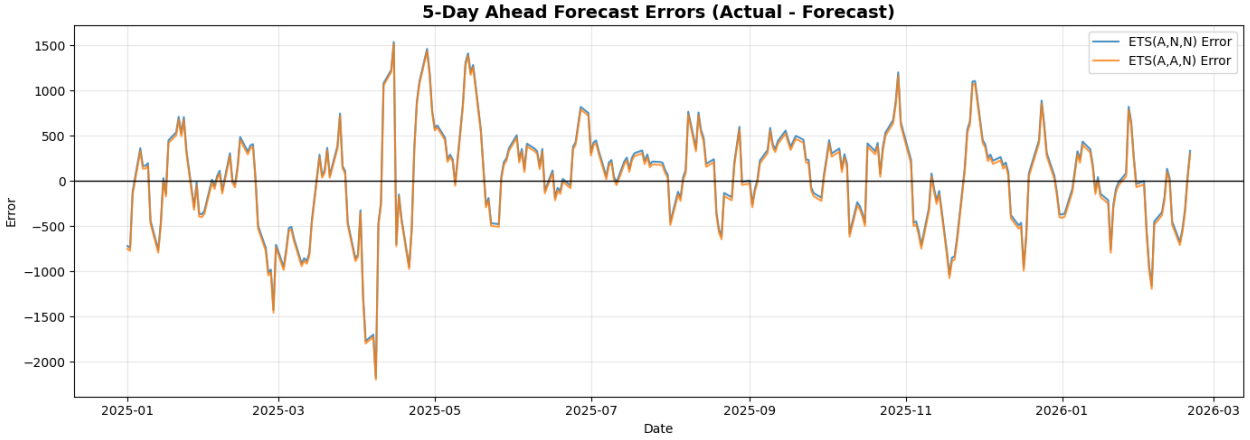
ARIMA



Date	Forecast (mean)
2026-02-23	22858.31
2026-02-24	22872.47
2026-02-25	22869.33
2026-02-26	22870.45
2026-02-27	22870.15

ETS





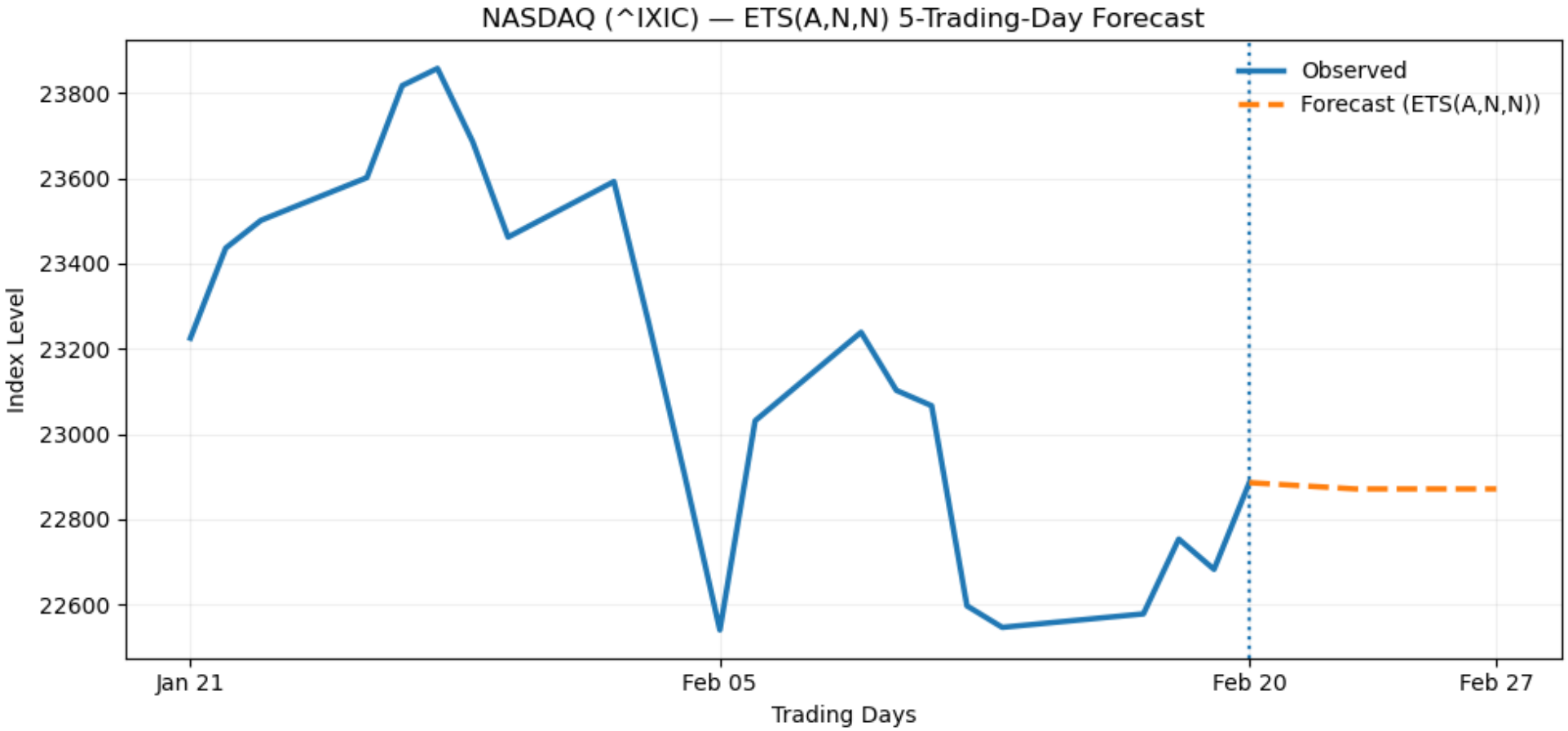
	Model	AIC
0	ETS(A,N,N)	26118.580772
1	ETS(A,A,N)	26151.273355
2	ETS(A,A,A) (s=5)	26199.654469

Best ETS model: ETS(A,N,N)

Final Model: ETS(A,N,N)

- Provides stable out-of-sample performance
- Avoids overfitting trend or seasonal components
- Simpler and more robust for financial time series
- Selected based on AIC

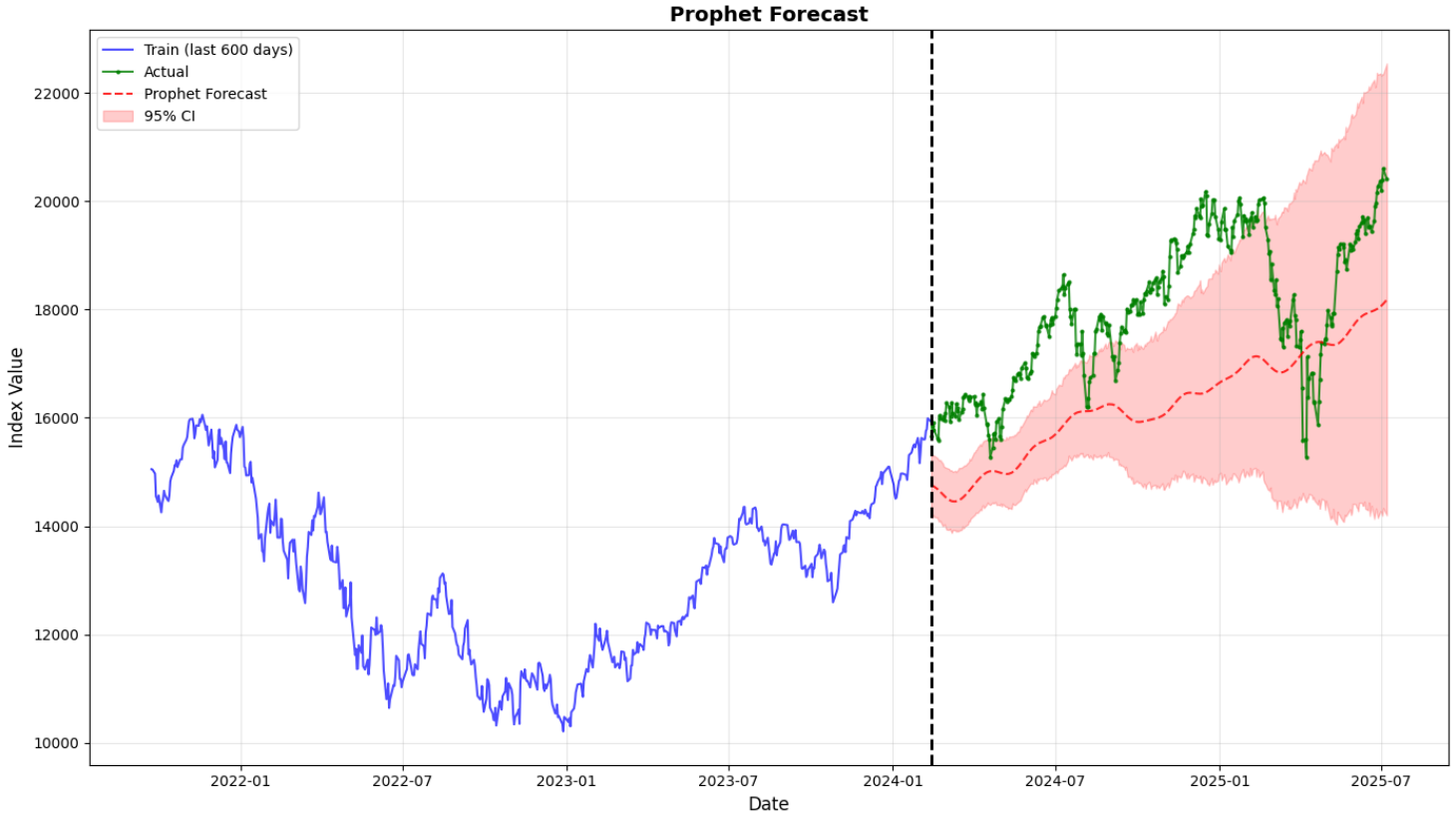
ETS



	Date	ETS_Forecast
0	2026-02-23	22871.33
1	2026-02-24	22871.33
2	2026-02-25	22871.33
3	2026-02-26	22871.33
4	2026-02-27	22871.33

Prophet – Long Term

- Assume yearly seasonality, no daily or weekly

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Mean Absolute Error (MAE): 1773.76
Root Mean Squared Error (RMSE): 1948.44
Mean Absolute Percentage Error (MAPE): 9.72%
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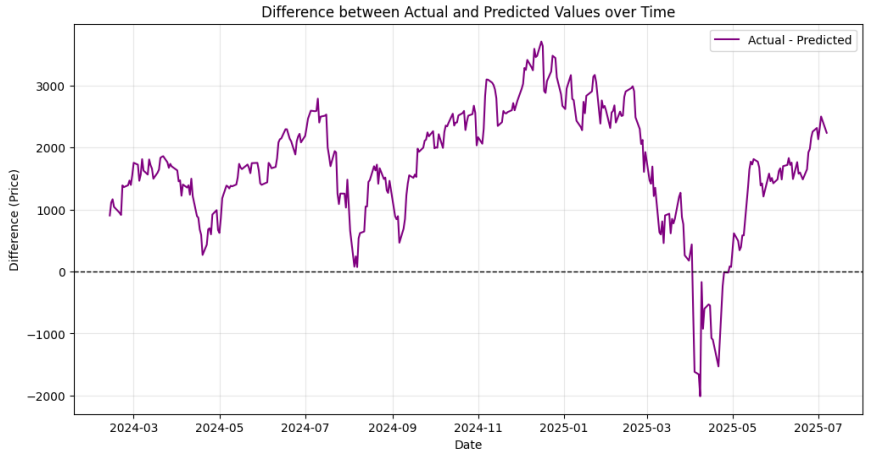
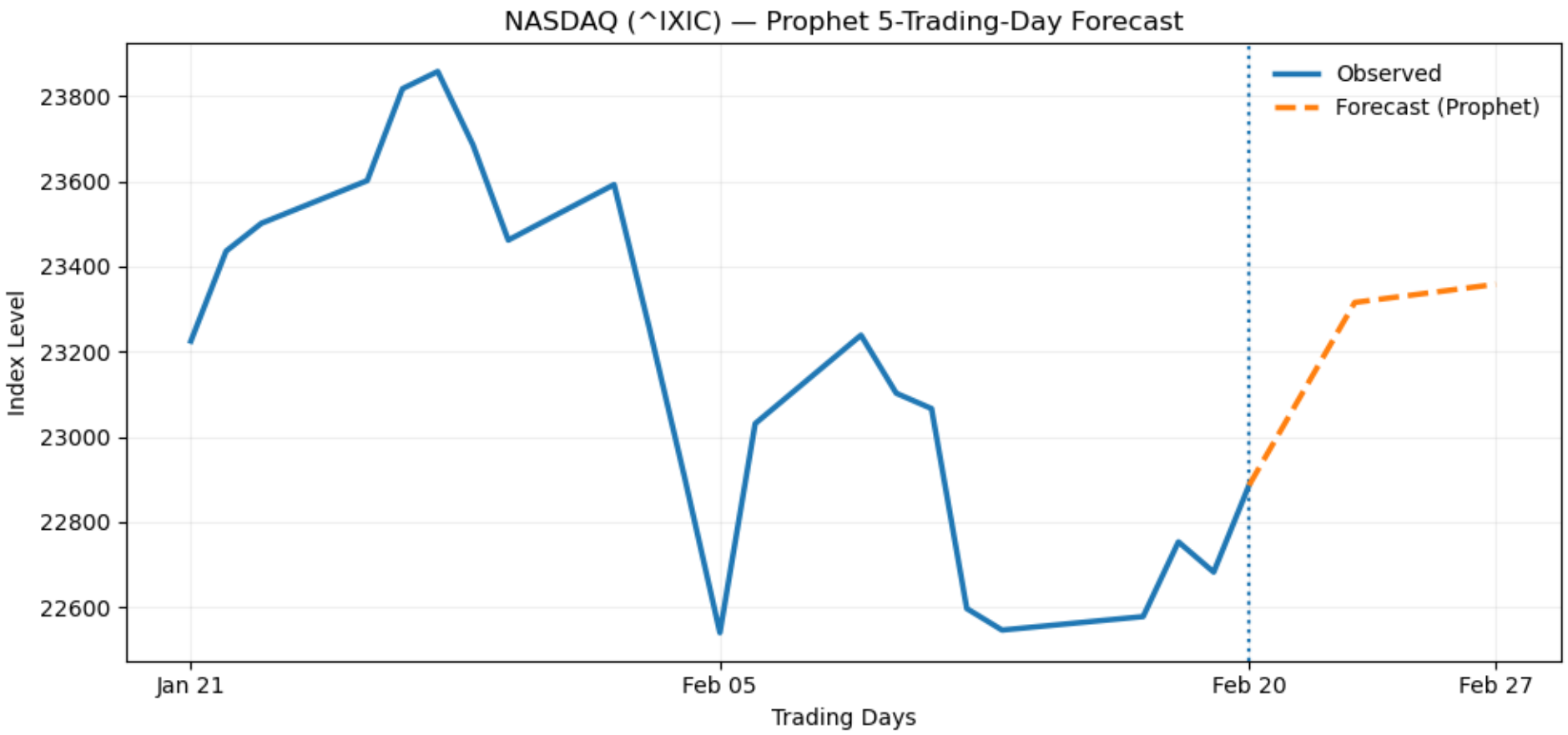
- Prophet model does a good job of tracking general data trend



Prophet



	Forecast_Date	Prophet_Forecast
0	2026-02-23	23315.510076
1	2026-02-24	23326.176691
2	2026-02-25	23336.843306
3	2026-02-26	23347.509921
4	2026-02-27	23358.176535

Forecast Evaluation Across Three Models

- We evaluate the models using out-of-sample forecasting.

Forecast accuracy was measured using:

MAE — average absolute error
RMSE — error penalizing large mistakes
MAPE — percentage error (scale-free)

5-day horizon
Train: 2016-01-04 → 2026-02-13 (n = 2640)
Test: 2026-02-16 → 2026-02-20 (n = 5)

1-month horizon (21 trading days)
Train: 2016-01-04 → 2026-01-22 (n = 2624)
Test: 2026-01-23 → 2026-02-20 (n = 21)

1-year horizon (252 trading days)
Train: 2016-01-04 → 2025-03-05 (n = 2393)
Test: 2025-03-06 → 2026-02-20 (n = 252)

Quantitative Comparison

5 Trading Days

Model	MAE	RMSE	MAPE
ETS(A,N,N)	**139**	**184**	**0.61%**
ARIMA(2,1,0)	160	202	0.70%
Prophet	711	723	3.14%

Over very short horizons, prices mostly follow the most recent level. Exponential smoothing adapts fastest to the last observed value, while trend-based models overreact.

1 Month

Model	MAE	RMSE	MAPE
ARIMA(2,1,0)	**469**	**551**	**2.05%**
Prophet	525	581	2.29%
ETS(A,N,N)	575	683	2.52%

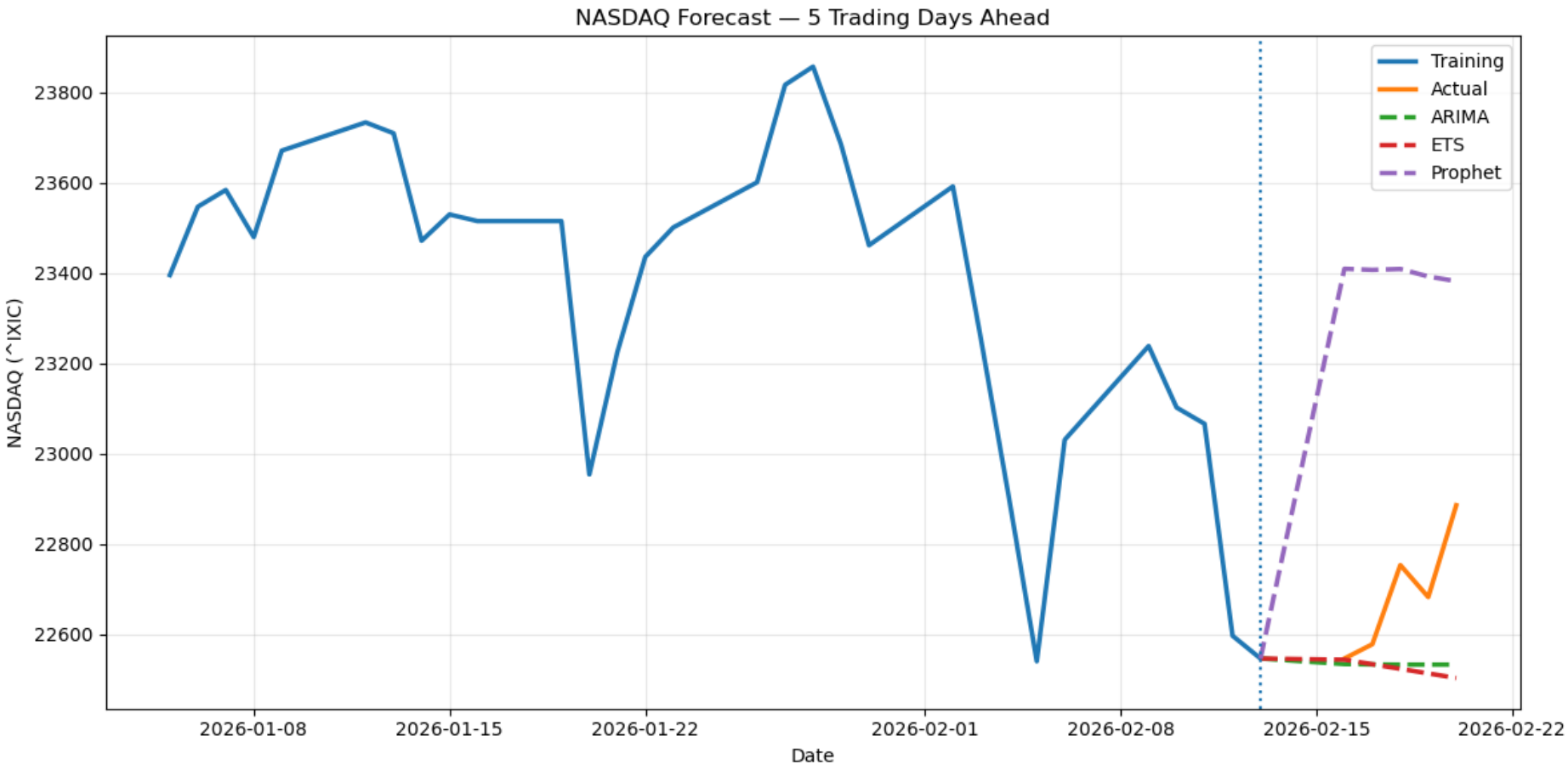
At this horizon the index shows short-term autocorrelation and momentum. ARIMA captures these dynamics through differencing and autoregressive structure, while ETS reacts too slowly and Prophet focuses on longer-term movement.

1 Year

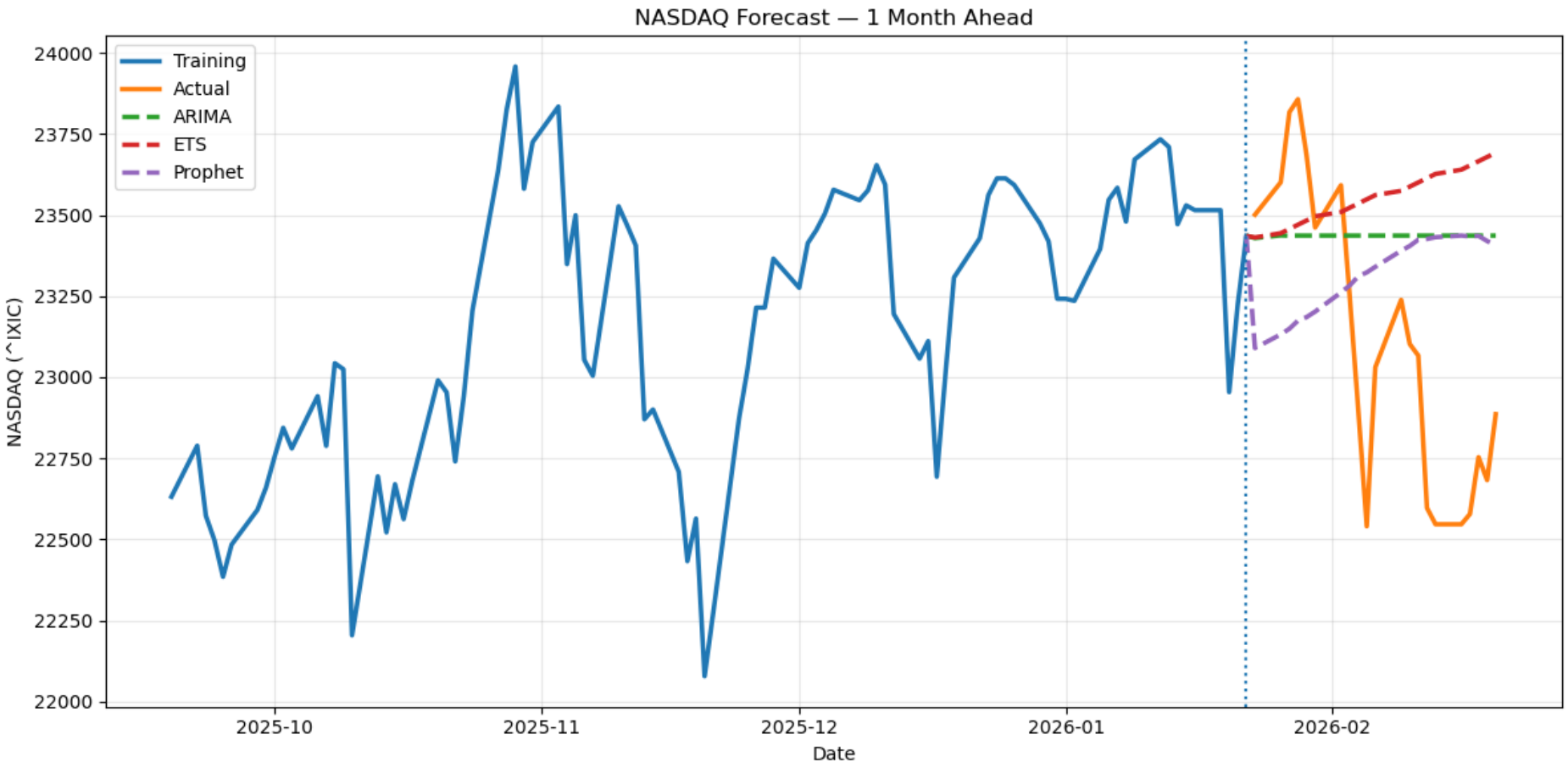
Model	MAE	RMSE	MAPE
Prophet	**1193**	**1608**	**6.31%**
ARIMA(2,1,0)	3020	3426	13.76%
ETS(A,N,N)	4833	5609	21.75%

Over long periods the index follows a gradual upward trend. Prophet models structural growth, while ARIMA accumulates forecast error and ETS lacks a trend component.

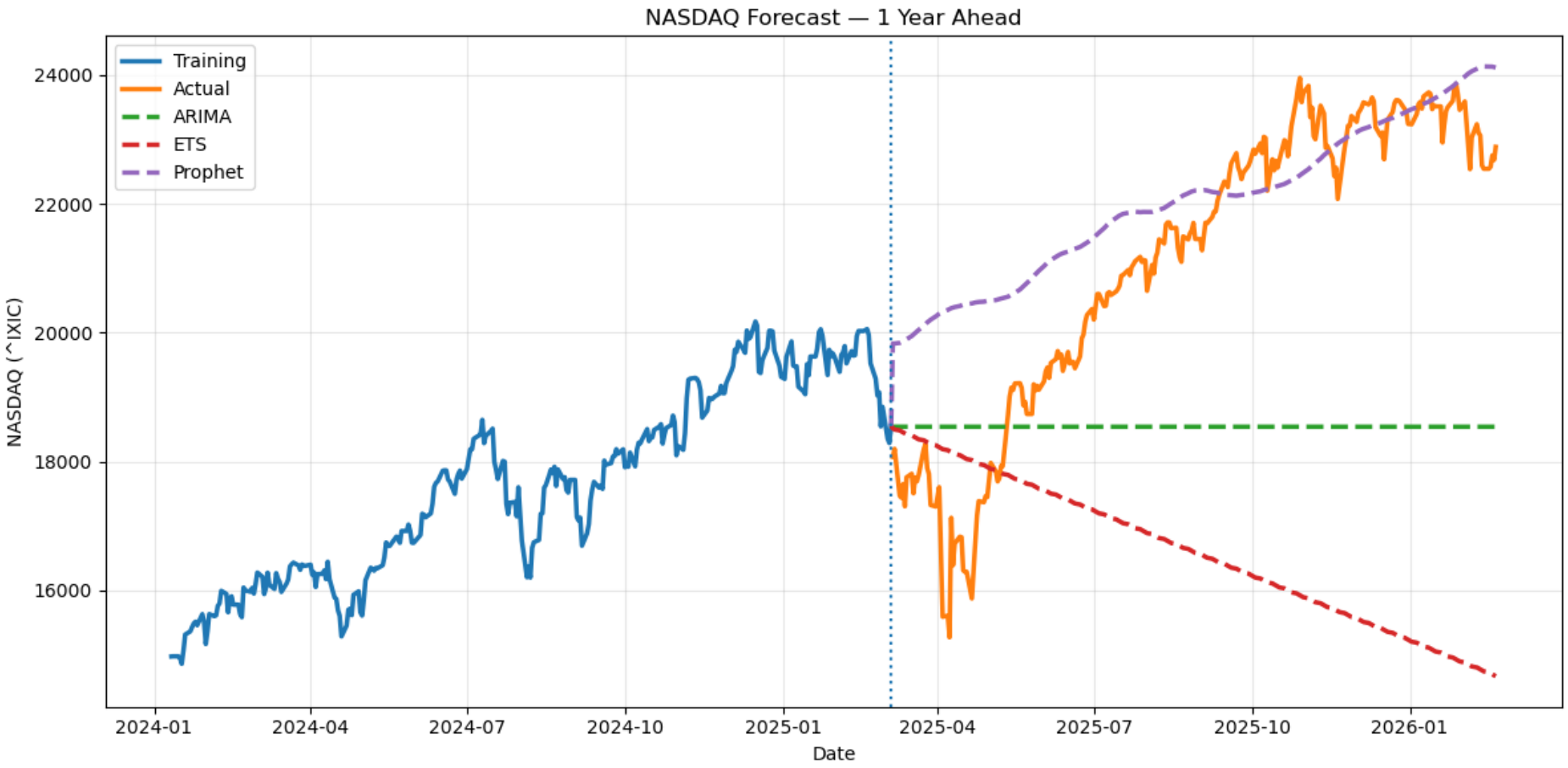
5 Day Ahead Comparison



1 Month Ahead Comparison



1 Year Ahead Comparison



Conclusion

- NASDAQ index is non-stationary and trend-driven.
- ARIMA effectively models short-term dependence after differencing.
- ETS(A,N,N) provides stable and robust short-term forecasts.
- Prophet performs well for long-term trend tracking.
- Forecast uncertainty increases with longer horizons.
- Financial time series remain inherently noisy and difficult to predict.